Network Influence Maximization Analysis Specification

Antoni Zyla Supervisor: Nathan Griffiths

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1 Abstract

A network is a weakly connected directed graph about which we reason. Influence is the measure of how ideas or properties spread through this graph based on certain models; choosing the seed set of nodes from which to begin the model determines how quickly and how far the ideas will propagate. Social networks [10], infectious diseases and behaviors [1] can be studied by using such an approach and can be used to inform decision making in many areas.

Finding a subset of nodes of the network which produce the greatest influence spread after a given time period could be used to determine those who'd best promote a product or be at greatest risk of spreading an infectious disease [1]. Finding this subset is difficult and known to be NP-Hard, as a reduction from Vertex Cover [7]. Using certain heuristics such as the degree of a node to choose the seed nodes is significantly more efficient running in polynomial time [3].

Using real world networks to perform this analysis on is not only difficult to obtain but also very large and computationally intensive [9]. I will aim to create a tool which can be used to evaluate and compare different heuristics and sampling methodologies against different types of networks. I will then use this tool to find a more representative sampling method on networks, or a heuristic which works more effectively on a certain set of topologies.

2 Problem Discussion

I will define a network upon which the models below will work on as a weakly connected directed graph. Doing so allows for modelling of networks in which the relationships are bidirectional (undirected) by simply creating two edges to represent that relationship, and for modelling directed networks such as email chains that have a clear sender and recipient. To be used for analysis, the network needs to be connected, as an idea cannot directly propagate between two disjoint components.

2.1 Models

1. Independent Cascade Model (ICM) - a node can be 'active' or 'inactive' and has a constant probability of infecting each of it's neighbours. Every infectious node has one attempt to infect it's neighbours before becoming inactive [7].

2. Decreasing Cascade Model (DCM) - an extension of the IC model the probability of infecting a node is determined by previous attempts, modelling how subsequent attempts are less likely to be successful after failed attempts [7].

3. Linear Threshold Model (LTM) - A model where each node has a certain 'threshold' required to infect it, considering the cumulative effect of the already infected nodes on the node determines if it will become infected [7].

2.2 Sampling Methods

Some real world datasets are far too large to efficiently compute these models on. Sampling these networks allows us to find a smaller network that encapsulates their features but is much more efficient to perform computations on. Those listed below I have researched and feel would be able to implement and use to evaluate against one another, as research progresses I may choose to add more methods or to not use a certain method.

1. Breadth First Search (BFS) - A natural exploration of the graph from a random node in layers [4].
2. Snowball Sample (SNS) - An extension of BFS which will stop once the desired sample size is reached [4].
3. Random Walk (RW) - Starting at a random node in the network chooses where to go onwards based on the the node's neighbours being chosen at random.
4. Metropolis Hastings Random Walk (MHRW) - A random walk with a modification to the probability function used to determine the subsequent node. This is to generate a uniform distribution of node degrees by favouring lower degree nodes [5].
5. Metropolis Hastings Random Walk with Delayed Acceptance (MHDA) - A further modification of MHRW in which the probability of backtracking to the previous node is reduced [8].
6. Albatross Sampling (AS) - A random walk with a chance of jumping to other locations in the network to find connected components [6].

2.3 Heuristics

Choosing which nodes to start the models on will affect how quickly the influence will propagate and how far it will spread to. A greedy approach may be close to optimal, however on larger datasets is inefficient and potentially intractable [2]. The following are some of the methods by which I will investigate, during the research phase I may add more or I may exclude some.

1. Random - As a baseline choose k random nodes.
2. Degree - Choose nodes with the highest degree.
3. Degree Discount - A modification of the degree heuristic but with a lower chance of choosing a node if it's neighbour is in the seed set [3].
4. Local Clustering Coefficient - The extent to which neighbours are connected to one another [4].
5. Average Neighbour Degree - Measuring the average degree of a node's neighbours may allow for communicating with high influence nodes [4].
6. Betweenness Centrality - Measuring how many shortests path's in the network pass throuhg a point [4].
7. Average Shortest Path Length - A measure of centrality by taking the mean length of the shortest path from the node to every other node [4].
8. Edge Embeddedness - For an edge (u, v) is the size of the intersection of the neighbourhoods of both u and v [4]. This serves as a measure of how tightly packed a network is around a certain area.
9. Edge Overlap - For an edge (u, v) the edge overlap is the ratio of it's embeddedness to the size of the union of the neighbourhoods of u and v. This can be used to determine the 'location' of an edge, for example if it connects connected components it may have a low overlap whilst a very high embeddedness may indicate not only is the agent well connected inside it's cluster it's also connected to other clusters [4].

10. Traditional Greedy Algorithm (Select Top k in [10]) - Choose greedily from nodes which have the highest influence [7]. This is inefficient due to requiring running a cascade for each node. It approaches within $(1 - 1/e)$ of the optimal solution so will be used to compare against heuristics.

2.4 Application to research

Utilising the above concepts, possible pieces of research in this area may include:

- Finding a more effective or representative heuristic for finding seed nodes in a network.
- Optimising a heuristic's running time whilst maintaining performance in regards to representation.
- Finding a more representative sampling method for varied network topologies.
- Finding a correlation between certain topological features and the effectiveness of one sampling method or heuristic over another, particularly in relation to that network feature.

3 Objectives

The exact goal of my project is not fully decided, however there are some requirements for the code. With these features I would be able to perform much of the computation and comparison of various papers. Ideally I would be able to find something new or improve on existing techniques as described in section 2.4.

1. Facilitate import and export of different graph formats.
2. Generation of synthetic graphs using graph generators for consistent analysis when real world datasets could be too large to run quick computations on.
3. Running and evaluation of the Independent Cascade Model, Decreasing Cascade Model and Linear Threshold Model. Allowing for evaluation of different heuristics against these models on different graphs.
4. Creation and evaluation of new sampling methods.
    - Evaluate the running time of sampling algorithms in practice, finding the real world constant factors.
    - Allow for evaluation of influence spread on the same network with different samples and how they affect it.
5. Creation and evaluation of new heuristics for selection of seed nodes against benchmark algorithms.
    - A 'cost' based strategy could be to give each algorithm a total cost, and each node has a value with the aim to find the subset that produces the greatest spread from nodes within a certain total 'cost'.
    - A 'minimum nodes' strategy to meet some spread metric, for example finding the lowest number of nodes that will produce a $\delta\%$ spread after x iterations.
    - Investigation of heuristics focused on overall growth in influence or rate of change in influence - this would involve charting the growth rate of various heuristics against eachother with trendlines etc.
6. Python **whl** and **tarball** packaging to allow for easy installation on various systems.
7. Programming with parallelisation in mind. Some results may take hours to compute on a personal device but are able to be ran in parallel on a cluster or more powerful system. I will attempt to use the batch compute system to produce any results.
8. Fully document all code, usage examples, parallelisation examples, and everything required to produce any results in my final report.

4 Methods and Methodology

The project will be broken up into three main stages; research, implementation, and evaluation. During each stage I will be in contact with my supervisor to ensure that I am on the right track and to keep me accountable.

1. The research stage will require a detailed literature review of articles and papers pertaining to network influence modelling, methods, heuristics and sampling. I will identify the validity and applications of each source and find whether there are any gaps in the presented knowledge to research further. I will repeat this process until my supervisor and I are happy the full scope of the project has been thoroughly reviewed.

2. The implementation stage of the project will require a methodology that can be adapted to changing requirements or a shift in focus based on information learnt in the later stages of the research phase. I will use an agile-like methodology, of one to two week sprints determined by meetings with my supervisor, with any changes to requirements determined during these meetings with future sprints based on the previous.

3. I will use my framework to replicate the results of the researched papers, showcasing how to create new sampling methods and heuristics, and how to compare the results of different sampling, heuristics and models on different networks.

4. Finally, I will attempt to improve upon a sampling method or heuristic and use my framework to do so. This will demonstrate that my tool is applicable to real world research uses and potentially find something interesting or noteworthy.

5 Timeline

The schedule is not fully decided at this point as the research phase is yet to be completed and the further points maybe change. Some points such as writing the report in the month leading to the deadline are certain, however the contents and direction of that report is not yet concrete. The purple bars on the gantt chart are the deliverables and the yellow bars are the main sub-tasks associated with those deliverables.

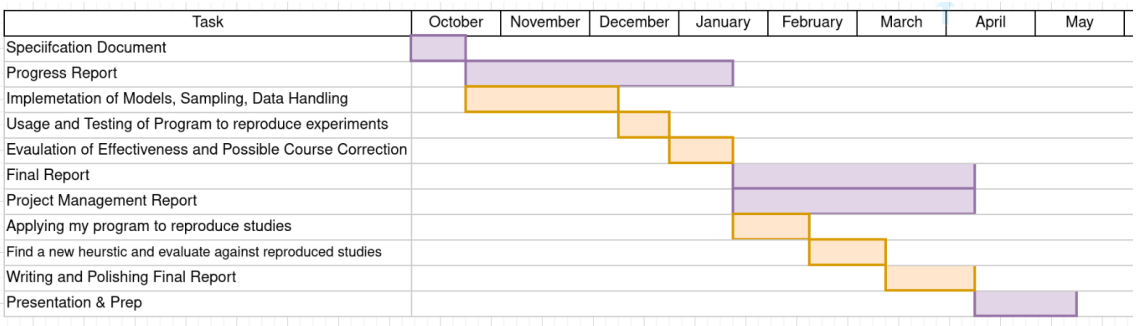


Figure 1: Gantt Chart of Expected Timeline

6 Resources and Risks

6.1 Resources

1. Python and Standard Libraries - NetworkX, NumPy, Pandas, Matplotlib, Build tools. These are all commonly available on a variety of systems, are open source, and can be built from source in case PyPI is unavailable.

2. Large Graph Datasets - Enron emails, airport traffic data, Gnutella data, Twitter tweet data and any other data sets as research progresses. The data is available from multiple sources on the internet, as well as being stored in the repository of my program under version control.

3. Batch Compute System - Department of Computer Science (DCS). The datasets I will be working with are large but not unmanageable on a typical desktop. The batch compute system will be used to learn and utilise good practice for scientific computing, as well as allowing for faster computation.

6.2 Risks

1. Failure to find a new heuristic or sampling method that is measurably better than those currently written about. In this case the project would still be a success as it would show that it allows for evaluation of heuristics effectively to be able to show that it is not statistically better.

2. Lost or damaged device containing code and documents. Using GitHub to be a remote mirror of local git repository and regularly committing code will prevent issues from lost work arising. DCS machines are more than adequate for processing needs.

7 Legal, Social, Ethical and Professional Considerations

At this stage there are no legal, social, ethical or professional considerations to take into account. The project will not be using, processing or interacting with personal data or information.

References

[1] J. Archbold, S. Clohessy, D. Herath, N. Griffiths, and O. Oyebode. An agent-based model of the spread of behavioural risk-factors for cardiovascular disease in city-scale populations. *PLOS ONE*, 19(5):1–22, 05 2024.

[2] J. Archbold and N. Griffiths. Limiting concept spread in environments with interacting concepts. In *16th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2017)*, pages 1332–1340. IFAAMAS, 2017.

[3] W. Chen, Y. Wang, and S. Yang. Efficient influence maximization in social networks. In *Proceedings of the 15th ACM SIGKDD international conference on Knowledge discovery and data mining*, pages 199–208, 2009.

[4] H. Franks, N. Griffiths, and S. S. Anand. Learning agent influence in mas with complex social networks. *Autonomous Agents and Multi-Agent Systems*, 28:836–866, 2014.

[5] M. Gjoka, M. Kurant, C. T. Butts, and A. Markopoulou. Practical recommendations on crawling online social networks. *IEEE Journal on Selected Areas in Communications*, 29(9):1872–1892, 2011.

[6] L. Jin, Y. Chen, P. Hui, C. Ding, T. Wang, A. Vasilakos, B. Deng, and X. Li. Albatross sampling: robust and effective hybrid vertex sampling for social graphs. 07 2011.

[7] D. Kempe, J. Kleinberg, and É. Tardos. Influential nodes in a diffusion model for social networks. In *international colloquium on automata, languages, and programming*, pages 1127–1138. Springer, 2005.

[8] C.-H. Lee, X. Xu, and D. Y. Eun. Beyond random walk and metropolis-hastings samplers: Why you should not backtrack for unbiased graph sampling, 2012.

[9] T. Wang, Y. Chen, Z. Zhang, T. Xu, L. Jin, P. Hui, B. Deng, and X. Li. Understanding graph sampling algorithms for social network analysis. *2011 31st International Conference on Distributed Computing Systems Workshops*, pages 123–128, 2011.

[10] W. Yang, L. Brenner, and A. Giua. Influence maximization in independent cascade networks based on activation probability computation. *IEEE Access*, 7:13745–13757, 2019.